Automatic Direction Finding System (ADF) – General Data

1. General

The Automatic Direction Finder (ADF) is used to display the aircrafts relative bearing to a tuned ground station.

For aircraft with S.O.O. 6388 incorporated, the Automatic Direction Finding (ADF) system is not installed.

For aircraft with S.O.O. 6388 incorporated, a single ADF and associated antenna are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6405 or a dual ADF and associated antennas are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6217.

2. Description

A. Automatic Direction Finder (ADF)

The ADF tunes frequencies from 19 kHz to 1,799 kHz and 2,180 kHz to 2,189 kHz. Functions consist of ADF, ANT, and BFO tuning modes. ADF 1 and optional ADF 2 antennas include both a loop and sense component and are installed at FS297.7 on the top of the fuselage.

The ADF system is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR) (refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data).

3. Adjustment/Test

A. ADF Function Test

1) General

These procedures are for the function test of ADF 1 and ADF 2.

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus), circuit breakers on the avionics circuit breaker panel are closed.

d) Within the ADF detail window, tune the ADF to local AM broadcast or a NDB station.

e) Select ANT mode on the ADF. Verify station audio is heard when ADF is selected on the audio control panel.

f) Select BFO mode on the ADF. Verify a 1 Khz tone is heard when the ADF is selected on the audio control panel.

g) Using the aircraft directional gyro or compass, turn the aircraft to the left 45 degrees. Verify ADF bearing error is not more than what is described in the manufacturers manual.

h) Continue to turn the aircraft stopping at 90 deg., 135 deg., and 270 deg. Verify ADF bearing error is not more than what is described in the manufacturers manual.

i) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – ADF Antenna

1) General

These procedures are for the electrical bonding testing for ADF 1 antenna and ADF 2 antenna.

2) Electrical Bonding Test

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the ceiling compartment forward of station 305.0.

c) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the ADF antenna backing plate. Place the second probe on the ADF antenna mounting screw anchor nut. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

d) Check the electrical conductivity between several random mounting screw anchor nuts within the electrical bond contact area to ensure consistency.

e) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

4. Removal/Installation

The ADF is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR), refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data, Para 4. for the MMDR Removal/Installation procedure.

A. Removal/Installation of the ADF 1 and ADF 2 Antenna (KA 44B)

1) General

These procedures are for the removal and installation ADF 1 and ADF 2 antenna (KA 44B).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the screws (2) from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the electrical connector.

f) Disconnect the electrical connector.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data). Note that the gasket is non-conductive and the skin does not need to be stripped. Bonding is through the grounding ring and mounting screws.

d) Bring the antenna assembly close to the fuselage, connect the electrical connector.

e) Secure the antenna assembly on fuselage with the mounting screws (2).

f) Solvent clean the exposed area of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

j) Check the antenna electrical bonding as per Para 3.B.

k) Remove safety clips and close MMDR (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel.

l) Do a function test on the NAV system as per Para 3.A.2).

5. Troubleshooting

The ADF is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR), refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data, Para 5. for the MMDR Troubleshooting procedure.

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